Entropy Rate $H(\mathcal{X}) = \lim_{n\to\infty} \frac{1}{n} H(X_1, \cdots, X_n)$

稳定 RP 的熵率必存在

Pf: $H(\mathcal{X}) = \lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n-1} H(X_{i+1} | X_i, \cdots, X_1)$

$H(X_{i+1} | X_i, \cdots, X_1) \leq H(X_{i+1} | X_i, \cdots, X_2) = H(X_i | X_{i-1}, \cdots, X_1)$ SSS

因 $H(X_i | X_{i-1}, \cdots, X_1) \geq 0$，单调有界 $\lim_{i\to\infty} H(X_i | X_{i-1}, \cdots, X_1)$ 存在

若 $a_i \to a$，则 $\frac{1}{n} \sum_{i=1}^{n-1} a_i \to a$ 故 $H(\mathcal{X})$ 存在

时不变马尔科夫链 $H(\mathcal{X}) = H(X_2 | X_1) = \sum_i \mu_i H(i)$

Pf: $H(X_1, \cdots, X_n) = \sum_{i=1}^{n-1} H(X_{i+1} | X_i) \underset{\text{Markov}}{=} nH(X_2 | X_1)$

e.g. $P = \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix}$，求 $H(\mathcal{X})$，$\lim_{n\to\infty} H(X_n)$

$\pi = \left(\frac{\beta}{\alpha+\beta}, \frac{\alpha}{\alpha+\beta} \right)$

则 $H(\mathcal{X}) = H(X_2 | X_1) = \pi_1 H(\alpha) + \pi_2 H(\beta)$

$= \frac{\beta}{\alpha+\beta} H(\alpha) + \frac{\alpha}{\alpha+\beta} H(\beta)$

$\lim_{n\to\infty} H(X_n)$ 即稳态 $H\left(\frac{\alpha}{\alpha+\beta}\right)$